

Lab Manual for Digital Logic and Design

LAB-01

Familiarization and verification of basic logic gates using ICs

EXPERIMENT NO. : 01

Familiarization and verification of basic logic gates using ICs.

Objective:

To become familiar with basic laboratory equipment and components. The basic logic gates are the basic building blocks of more complex logic circuits. The purpose of this lab is to learn about Digital Logic and digital logic circuits. By the end of this lab you will have an understanding of the digital electronics, functions and operations of different logic gates.

Apparatus:

- Logic trainer/ MultiSim Software
- 14 pin ICs(7408,7400,7432,7402,7486,7404)
- Power supply

Theory:

PART (A)

DC Power Supplies:

A Direct Current (DC) power supply is a source of voltage that provides ideally a fixed terminal voltage even though the current demands from the electrical system may vary. A variable voltage-regulated supply is used in the laboratory for performing experiments. This supply can be manually adjusted to deliver any required voltage within its range of operation. The word voltage-regulated means that the voltage delivered by the supply remains constant despite changes in load current, within specified limits. Some power supplies are designed to provide two or more independent dc voltages; in which case the instruments have separate controls and separate output terminals. The polarity of the DC terminals on the supply is usually marked either -, +, or GND. By convention a red jack is used for positive terminal and a black jack is used for the negative terminal of the supply.

How to use a Power Supply:

The following general precautions should be observed in using a power supply:

1. Never create a short circuit across the output terminals, or you may damage the supply. Keep the leads connected to the output terminals from making contact with each other.
2. If any component on the experimental circuit appears to be overheating after power is applied, turn the supply off and determine the cause before you proceed with the experiment.
3. To prevent damage to the power supply do not switch the supply on and off excessively. If an experiment requires power to be interrupted frequently, use an external switch on the bread board to apply and remove power from the circuit.
4. Do not operate the power supply beyond its rated current capacity. If an ammeter indicates that you are exceeding the current capability of the power supply, turn it off and check the experimental circuit to determine why it is drawing excessive current.

Instruments:

Electrical and electronic instruments are used in the laboratory to measure quantities, to provide external signal stimuli to circuits, and to view graphically the behavior of circuits.

Digital Multi-meter:

DMM can be used to measure AC & DC voltages and currents, resistance, frequency. Many DMMs can also be used to test transistors and diodes. When measuring voltage, the first step is to correctly configure your probes. Plug the positive lead (red) into **VΩmA** and the negative lead (black) into **COM**. The next step is Powering on the DMM; the **Power** button is found on the middle under the multi range selector switch. To measure DC voltage, rotate the multi range selector switch to **DC V** range after correctly configuring the DMM as instructed above. The DMM is able to measure a wide range of DC voltages; however, it must be set to the correct range in order to return a good reading. These ranges are 200mV, 2000mV, 20 V, 200 V, 1000 V etc. To measure AC voltages rotate the range selector switch to **AC V** range after correctly configuring the DMM as instructed above. All measurements of AC signals are root mean square (RMS) values. When measuring AC voltages it is also essential that your DMM be in the correct range in order to return a good reading. The voltage range in the AC mode is 200V, 750V (rms value).

Bread Boards Basics:

Breadboards are simply a set of pre-wired interconnects that aid you in the building of your circuits. By plugging a wire of one component into a hole you will connect it to all other components in that strip or bus. Using strips, buses, and jumper wires you can construct a circuit on your breadboard. You can tell which holes are connected in one node by the black lines connecting them in Figure 1.1. There are the component connection strips that run up and down and buses labeled with an A or B.

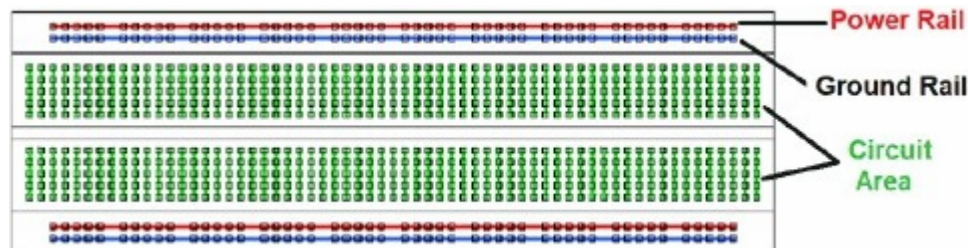


Figure 1.1: Breadboard

Digital electronic trainer:

The M21-7000 DIGITAL LOGIC LAB is a comprehensive and self-contained system suitable for anyone engaged in digital logic experiments. All necessary equipment for digital logic experiments such as power supply, signal generator, switches and displays are installed on the main unit. The modules cover a wide variety of essential topics in the field of digital logic. It is a time and cost saving trainer for both students and engineers interested in developing and testing circuit proto- types.



Figure 1.2: Digital electronic trainer

- Suitable for combinational logic, sequential logic and microprocessor circuits design and experiments.
- Ideal tool for learning the basics of digital logic circuits.
- Comprehensive power, signal supply and testing devices for convenient experiments.
- Experiments are flexible with universal breadboard.
- Capable of processing TTL, CMOS, NMOS, PMOS and ECL circuits.
- All supply units are equipped with overload protection for safety purpose.
- All modules equipped with 8-bit DIP switch for fault simulations.
- Individual keeping cases for all modules for easy storing and carrying.

PART (B)

Integrated Circuits and Gates:

Digital circuits are hardware components that are implemented using transistors and interconnections in complex semiconductor devices called integrated-circuits. Digital circuits work in binary logic domain which uses two discrete values, TRUE (High) and FALSE (Low). We can also refer to these values as 1(High) and 0 (Low).

The logic gate is a digital circuit that is able to operate on a number of binary inputs in order to perform particular logic function. The total number of possible combinations of binary inputs to gate is determined by $N=2^n$

Where N =number of possible input combination
 n =number of input variables

Examples of the basic logic gates are AND, OR, NOT, NAND and NOR. A complex gate such as an XOR gate can be built out of these basic gates. Each logic gate is actually implemented in part of an integrated circuit (IC), with each gate made using several transistors. For the most part we do not need to concern ourselves with the actual

circuitry inside each gate, only the interconnections between individual gates. Usually each IC package contains several individual gates. The 7408 chip implements 4 two input AND gates and is commonly referred to as a “Quad two input AND” chip. Similarly the 7432 chip is a “Quad 2 input OR” chip while the 7404 chip is a “Hex Inverter” since it contains 6 inverters. The pins are numbered as shown in Figure 1.3. Pin 1 is usually identified as the pin to the left of an indentation or cutout in one end of the chip that is visible when the chip is viewed from the top. Occasionally, it is also identified by a printed or indented dot placed just next to it.

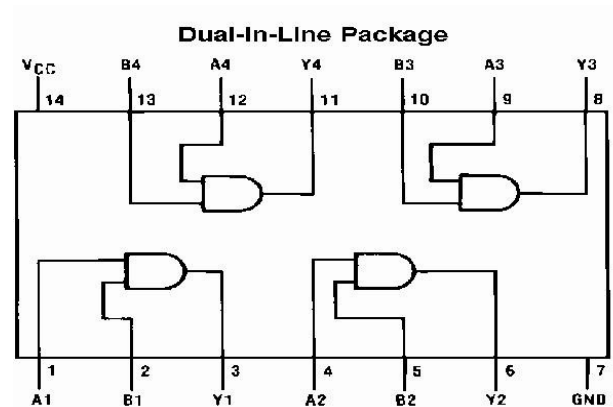


Figure 1.3: Numbering of pins

Figure 1.3, shows an IC which contains four 2-input AND gates. You will notice that pins 7 and 14 show no connections. In 14 pin DIP packages, pin 7 is usually connected to ground (Gnd), and pin 14 is usually connected to the power supply (Vdd). These connections must be made or the chip will not work.

A few of the more common TTL "DIP" circuit packages are shown in figure 1.4 for reference:

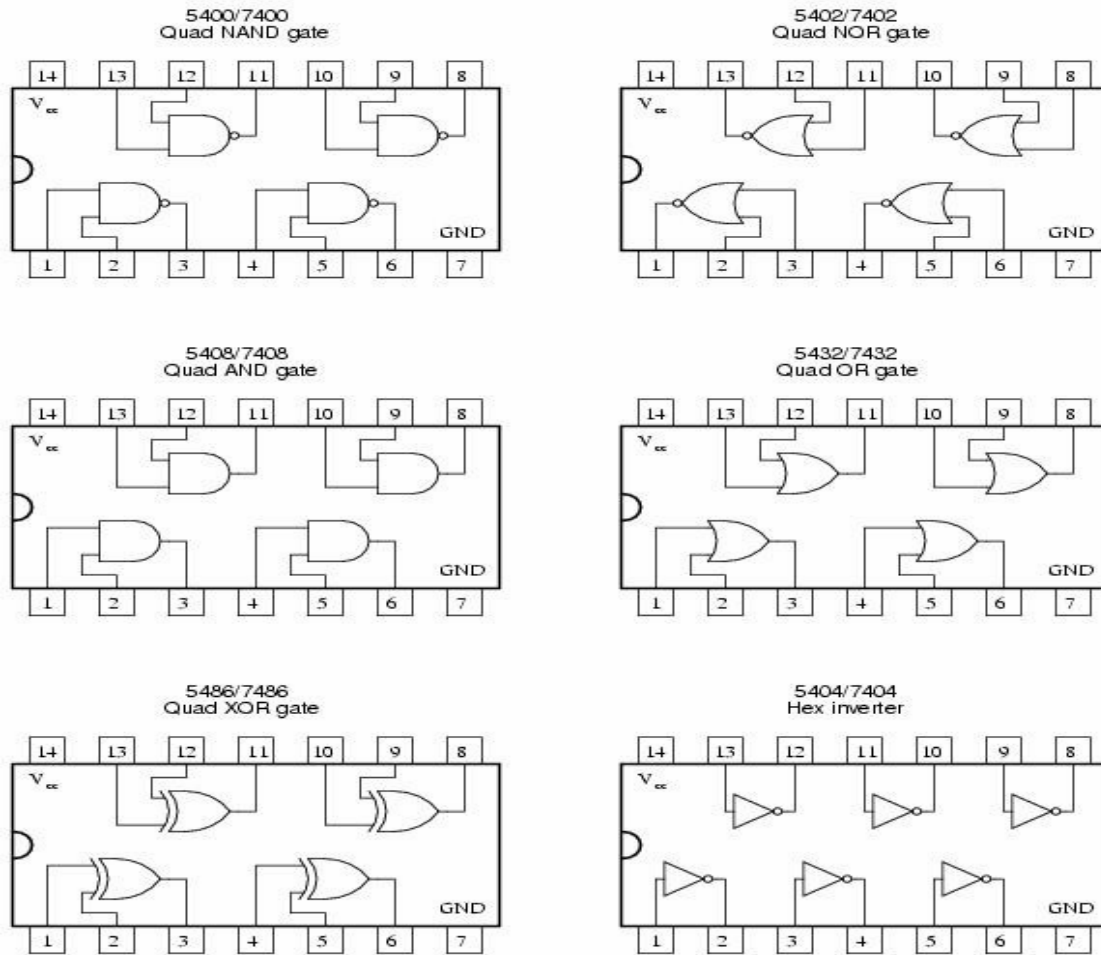


Figure 1.4

AND Gate

The AND function is similar to the multiplication in mathematics. This is the all or nothing operator and it provides a logic 1 output only when all the inputs of the gate are at logic 1, and logic 0 output for all other input combinations. The AND function is described in terms of the following “truth table”. In terms of logic symbols, $OUT = A \cdot B$

OR Gate

The OR function is similar to the mathematical function of addition and the output for the OR gate may be analyzed using the laws of addition. The logic operator for the OR function is a + sign. The output will be logic 0 only if all the inputs are logic 0, and the output will be logic 1 anytime any input is at logic 1. Here, $OUT = A+B$.

NOT Gate

The NOT circuit or inverter performs the basic logic function of complementation. It may be identified by the presence of a bubble on the input or the output of the traditional logic symbol. The output of NOT gate is the inverse of the input. Unlike the others it only has one input and one output.

NAND Gate

The NAND function is the complement of the AND function and the logic symbols have the inversion on the output. NAND gate is constructed by adding an inverter after AND operator. The NAND function provides logic 0 on the output only when both inputs are logic 1, and logic 1 output for all other combinations. Here, $OUT = \overline{A.B}$

NOR Gate

The complement of the OR function is the NOR function and the logic symbol has the inversion present on the output. NOR gate is constructed by adding an inverter after OR operator. Here, $OUT = \overline{A + B}$

Procedure:

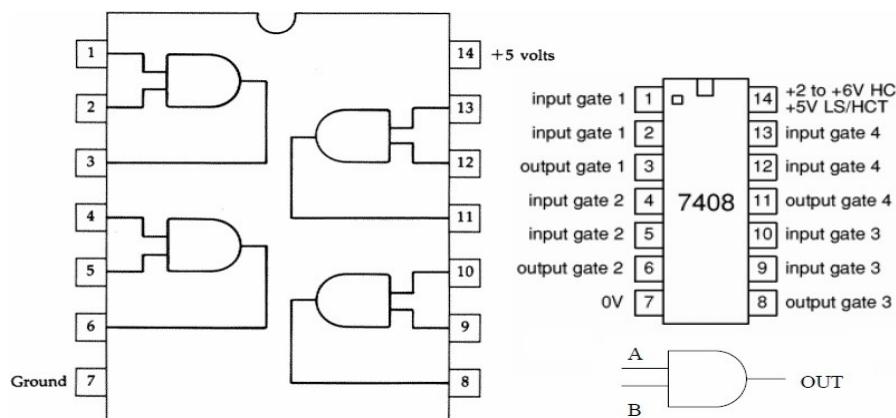
- Place the IC 7408LS on the trainer board.
- Connect V_{cc} and ground to the respective pins on the trainer board.
- Connect the inputs to the input switches provided in the trainer board.
- Connect the outputs to the switches of output LEDs.
- Apply various combinations of inputs as shown in the truth table and observe the conditions of LEDs.
- Similarly follow these steps for other gates.

Observations & Calculations:

AND Gate:

Truth table for IC -----74LS08D-----

Input 1	Input 2	Output
0	0	0
0	1	0
1	0	0
1	1	1



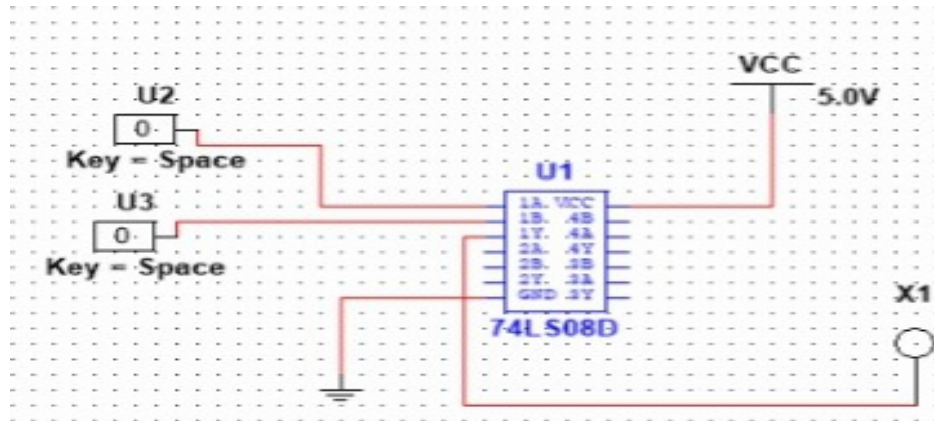


Fig: Simulation AND-Gate

OR Gate:

Truth table for IC ---74LS32D-----

Input 1	Input 2	Output
1	1	1
1	0	1
0	1	1
0	0	0

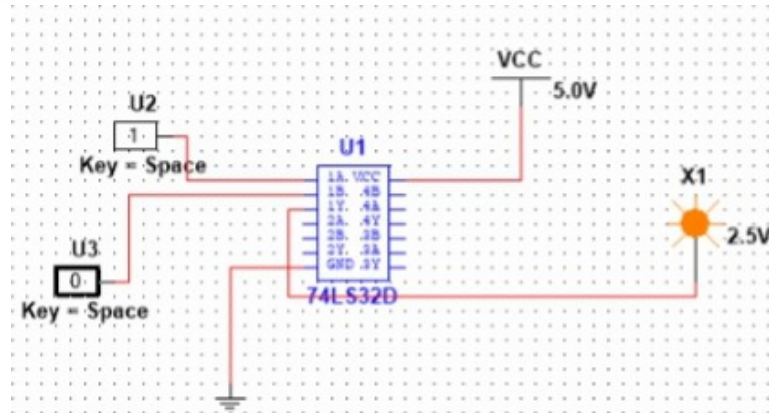
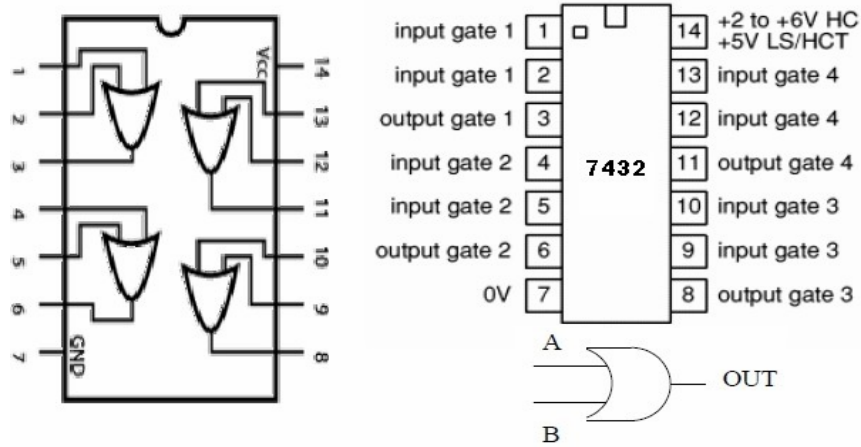


Fig: Simulation OR-Gate

NOT Gate:

Truth table for IC ---74LS04D-----

Input	Output
1	0
0	1

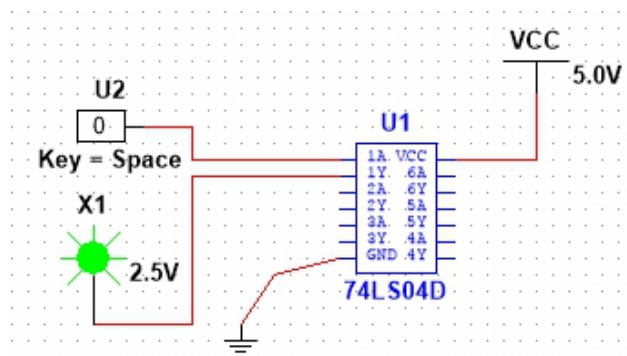
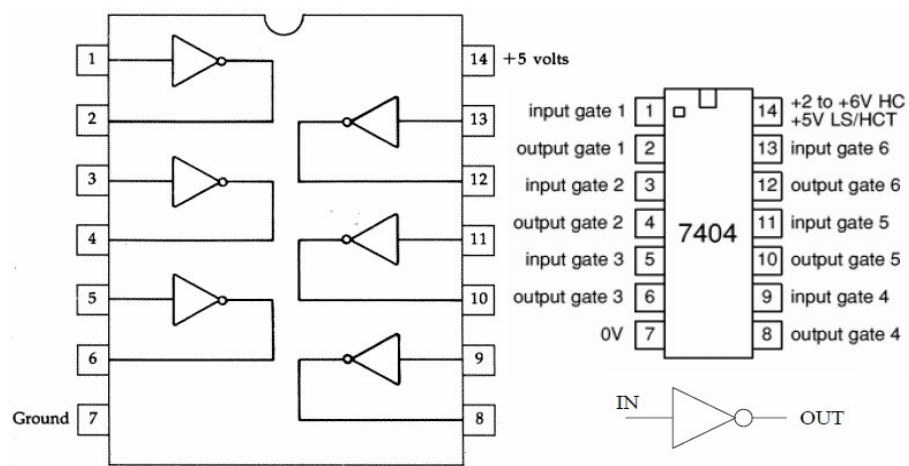
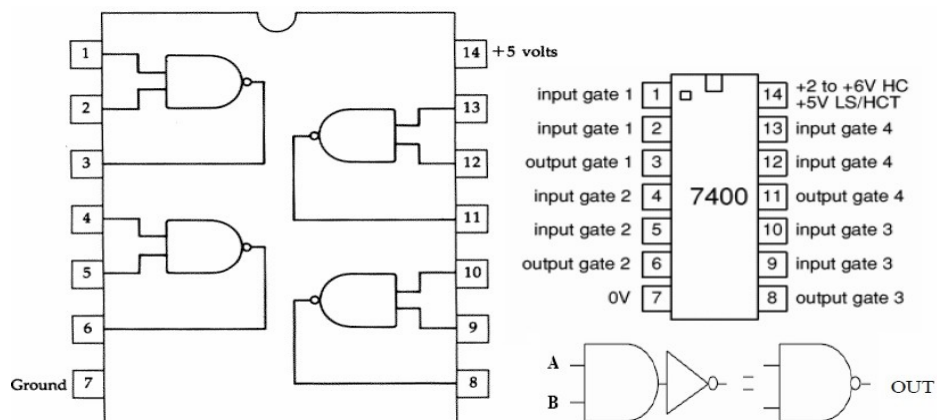


Fig: Simulation Not-Gate

NAND Gate:

Truth table for IC ---74LS00D---

Input 1	Input 2	Output
0	0	1
0	1	1
1	0	1
1	1	0



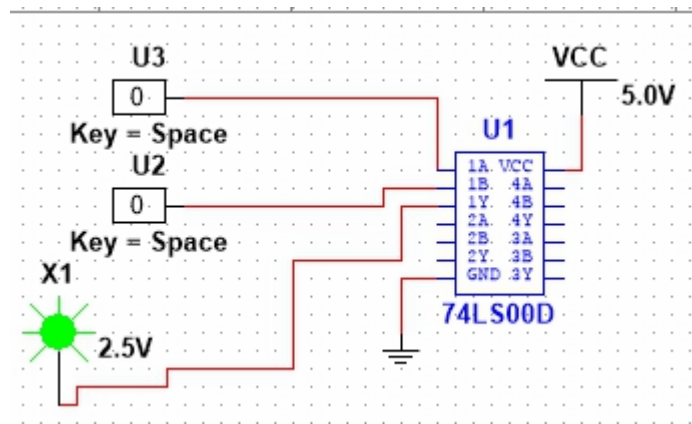


Fig: Simulation NAND-Gate

NOR Gate:

Truth table for IC -----74LS02D-----

Input 1	Input 2	Output
0	0	1
0	1	0
1	0	0
1	1	0

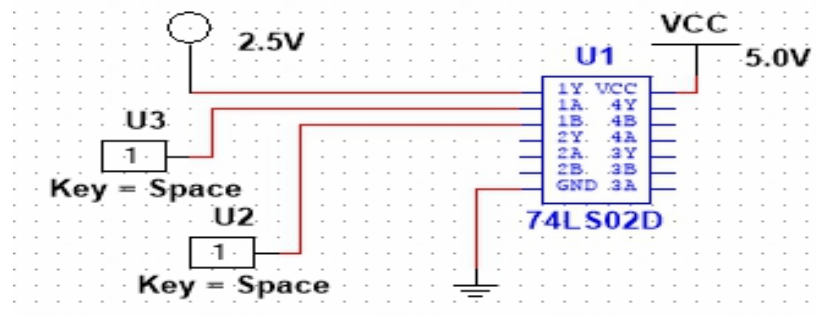
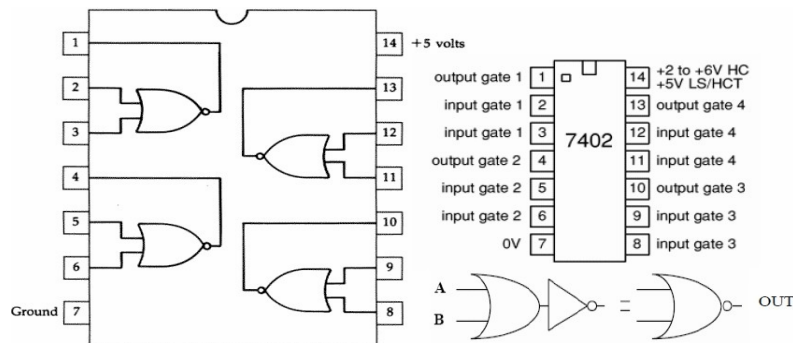


Fig: Simulation NOR-Gate

Conclusion:

Implemented the following logic gates in MultiSim and the outputs were observed on LED's.